Kunwoo Lee

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Education

Master of Science in Mechanical Engineering – Research May 2026
Carnegie Mellon University, Pittsburgh, PA
Bachelor of Science in Mechanical Engineering, *Summa Cum Laude* May 2024
Illinois Institute of Technology, Chicago, IL
Bachelor of Science in Mechanical Engineering, *Honor Student* Feb 2017 – Dec 2021
Inha University, Incheon, South Korea

Master's Research Experience & Projects

Sparse-Sensor Motion Tracking for Accessible Biomechanics Aug 2025 – Present
Graduate Research Assistant & MS Research (24-794), Carnegie Mellon University, PA
- Developing a generative AI system to recover full-body kinematics from sparse and accessible sensors (e.g., VR headset and controllers), with potential extensions to RGB or IMU inputs, with generalizability to out-of-distribution single-legged tasks.

Calibration-Free Multimodal Physical Therapy (PT) Assessment May 2025 – Aug 2025
Graduate Research Assistant, Carnegie Mellon University, PA
- Implemented a computer vision pipeline that autocalibrates two RGB cameras and aligns uncalibrated IMU signals to body segments, removing manual setup and enabling scalable PT assessment.
- Trained a random forest classifier on kinematic outputs from our auto-calibrated, multi-view fusion system; the model exceeded the clinician agreement benchmark (66.8%), demonstrating performance at the level of expert consensus.

Validating Computer Vision-based Human Kinetics Estimation Sep 2024 – Aug 2025
Graduate Research Assistant & MS Research (24-794), Carnegie Mellon University, PA
- Evaluated commercial and open-source softwares for human kinetics estimation against the gold-standard marker-based approach, enabling more informed decisions for the trade-off of accessibility and precision.
- Introduced a novel validation framework by simulating marker registration uncertainty to quantify variability in marker-based methods, demonstrating that state-of-the-art multiview kinetics estimation can achieve comparable accuracy under specific conditions.

sEMG-Based Keyboards-Free Typing with Deep Learning Jan 2025 – May 2025
Intro to Deep Learning (11-785) Course Project, Carnegie Mellon University, PA
- Developed a transformer-based model that maps wrist sEMG signals to keyboard input, improving prediction accuracy by 25% compared to leading Time-Depth Separable Network baseline.
- Conducted large-scale deep learning experiments across HPC resources (PSC Bridges-2) using Slurm, ensuring efficient, reproducible training with PyTorch Lightning and Hydra.

Undergraduate Research Experience & Projects

Equity Analysis of Tax Increment Financing (TIF) Jan 2024 – May 2024
Interprofessional Projects (IPRO 497), Illinois Institute of Technology, IL
- Uncovered disparities in the economic impact of TIF, showing disproportionate benefits for White, higher-income wards through comparative analysis of TIF and non-TIF districts across racial and income groups, using clustering methods.
- Presented findings to peers and faculty, sparking discussions on the equity implications of municipal finance policies in urban planning.

CNN-Based Estimation of Airfoil Kinematics Jan 2023 – May 2024
Undergraduate Research Assistant, Illinois Institute of Technology, IL
- Designed and trained a convolutional neural network (CNN) using TensorFlow to predict airfoil pitching kinematics from instantaneous downstream wake snapshots, offering faster prediction from smaller measurement fields, achieving over 90% accuracy.
- Generated high-fidelity data using direct numerical simulation (DNS) of both periodic and non-periodic airfoil motions, enabling robust training and evaluation of the CNN model across diverse flow regimes.

AI & Computational Modeling for Healthcare Applications Jan 2022 – Jul 2022
Undergraduate Research Assistant, Inha University, South Korea
- Achieved 98.6% accuracy in classifying five arrhythmia types from ECG signals—surpassing the state-of-the-art benchmark

(97.3%)—using a CNN model built in TensorFlow.

- Quantified how anastomosis angle and graft porosity affect vascular bypass patency using computational fluid dynamics, showing that larger angles reduce low wall shear stress regions and may lower restenosis risk.

Publications and Posters

- **Kunwoo Lee***, Soyong Shin*, Keelan Enseki, Adam Popchak, Jessica Hodgins, Eni Halilaj, *Domain Knowledge Improves AI-Based Evaluation of Physical Therapy Performance.* (In preparation) **[Manuscript] (* Equal contribution)**
- Chaeeun Lee*, **Kunwoo Lee***, Zhixiong Li, Vu Phan, Evy Maeinders, Eni Halilaj, *Does Markerless Motion Tracking Hold Up Under Load? Evaluation of Multi-View Methods for Lower-Extremity Joint-Moment Estimation.* (In preparation) **[Manuscript] (* Equal contribution)**
- Zhixiong Li, Chaeeun Lee*, **Kunwoo Lee***, *Videos Capture Functional Strength Deficits Following ACL Reconstruction.* American Society of Biomechanics, 2025 **[Conference Poster] (* Equal contribution)**
- **Kunwoo Lee**, Youngjo Choi, Lane Hartwig, Nam Gyu Lee, Juhyun Jung, Andres Iniguez, Zaid Iqbal, Darshan Sasidharan Nair, Jean-Yves Thompson, *Understanding the Varied Economic Impact of Tax Increment Financing (TIF)*. Illinois Institute of Technology Innovation Day, 2024 **[Poster Presentation]**
- **Kunwoo Lee**, Scott T.M. Dawson, *Estimating Pitching Airfoil Kinematics Using Convolutional Neural Networks.* Illinois Institute of Technology MMAE Poster Competition, 2023 **[Poster Presentation]**
- **Kunwoo Lee**, SeokJu Lee, Woodam Kim, Kyung Eun Lee, *The Effects of Anastomosis Angle on Blood Flow in Bypassed Blood Vessel.* Biomedical Engineering Society (Circulation), 2022 **[Conference Poster]**

Graduate Coursework

Intro to Deep Learning (11-785)
Biomechanics of Human Movement (24-663)
Robot Dynamics and Analysis (24-760)
Learning for 3D Vision (16-825)
Computer Vision for Engineers (24-678)

Skills

Programming:	Python, C++	**ML/AI Frameworks:**	PyTorch, TensorFlow
Cloud/HPC:	Slurm, AWS	**Biomechanics Tools:**	OpenSim, Theia3D, Visual3D, OpenCap
Visualization:	AitViewer, PyTorch3D		

Professional & Leadership Experience

Armour Chapter, Triangle Fraternity Jan 2023 – May 2024
Illinois Institute of Technology, Chicago, IL

- Directed alumni engagement initiatives through event planning (annual dinner, reunions, fundraising) and by editing the Triangle magazine to communicate chapter activities.

Private Tutor Mar 2020 – Jul 2022
Incheon, South Korea (Part-time, 3 hrs/week)

- Provided individualized math instruction and mentoring, strengthening conceptual understanding and problem-solving; student improved from C to A over two years.

Unit Supplier & Squad Leader Jan 2018 – Sep 2019
ROK Army 2nd Corps, Chuncheon, South Korea

- Oversaw food supply logistics for unit operations, including ordering, inspection, transport, and distribution.
- Led and counseled squad members, fostering morale and teamwork through structured team-building activities.

Awards & Honors

Summa Cum Laude	Illinois Institute of Technology	2024
Dean's List	Illinois Institute of Technology	2022–2023
TOEFL Exchange Student Scholarship	ETS Korea	2022
Honor Student	Inha University	2020–2021
Academic Excellence Scholarship	Inha University	2020–2021
Soldier of Valor Award	Republic of Korea Army	2019